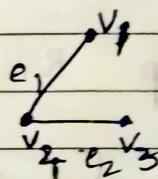
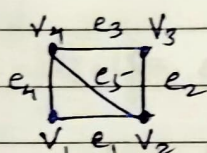
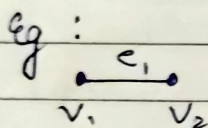


## MODULE - 3

## GRAPH THEORY

### Graph

A Graph  $G$  is a mathematical structure consisting of 2 sets  $V$  &  $E$  called the set of vertices & set of edges.



### Basic Ideas of Graph

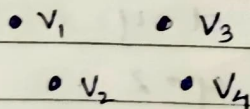
#### 1) Trivial Graph :-

A graph consisting of only 1 vertex & no edges.



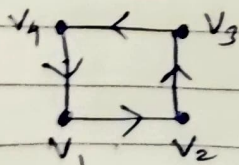
#### 2) Null Graph :-

A graph consisting of  $N$  vertices & no edges.



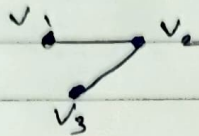
#### 3) Directed Graph :-

A graph consisting of the direction of edges is called directed graph.



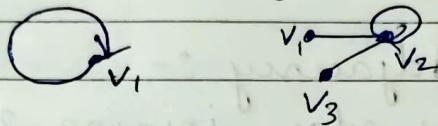
4) Undirected Graph:-

A graph which is not directed.



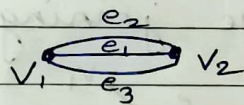
5) Self loop in a Graph:-

If edges having the same vertex



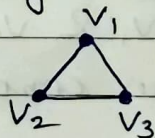
6) Multiedge:-

A collection of 2 or more edges having the same end points.



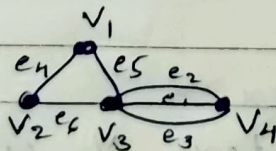
7) Simple Graph:-

A graph doesn't contain any self loop & multiedge.



8) Multigraph:-

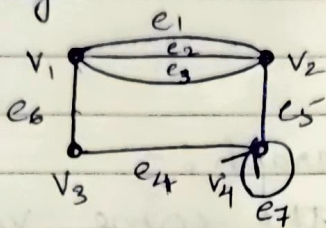
A graph doesn't contain any self loop but contains multiedges.





9) Pseudo Graph :-

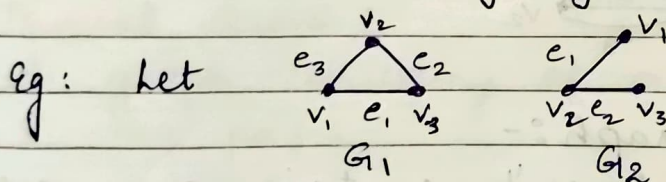
A graph that contains both self loop & multiedges.



10) Incidence & Adjacency :-

Let  $e_i$  be any edge joining 2 vertices  $v_i$  &  $v_j$ , then  $e_i$  is said to be incident of  $v_i$  &  $v_j$ .

2 vertices are said to be adjacent if there exist 2 edge joining the vertices.



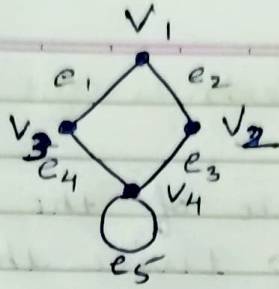
In graph  $G_1$ ,  $e_1$  is incident to  $v_1$  &  $v_3$ . &  $v_1$  &  $v_3$  are adjacent because they are connected by  $e_1$ .

In graph  $G_2$ ,  $e_1$  is incident to  $v_1$  &  $v_2$ , but  $v_1$  &  $v_3$  are not adjacent because there is no edge.

11) Degree of the Vertex :-

The degree of a vertex  $v$  in a graph  $G$  denoted by  $d(v_i)$  is equal to the no. of edges which are incident to  $v_i$  with self loop counted twice.

Eg:



$$d(V_1) = 2$$

$$d(V_2) = 2$$

$$d(V_3) = 2 + 2 = 4$$

$$d(V_4) = 2 + 2 = 4$$

## 12) Isolated Vertex & Pendant Vertex :-

A vertex having degree 0 is called isolated vertex.

A vertex having degree 1 is called pendant vertex.

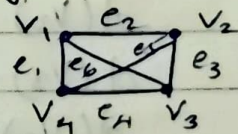
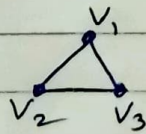
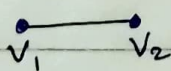
## 13) Finite & Infinite Graph :-

A graph with a finite no. of vertices as well as edges is called finite graph, otherwise infinite graph.

## 14) Complete graph :-

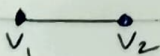
A complete graph is a graph if each vertex is connected to every other vertex.

Eg:

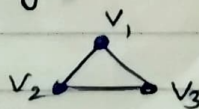


## 15) Regular graph :-

A graph  $G$  is regular if every vertex have same degree



1-regular graph



2-regular graph.



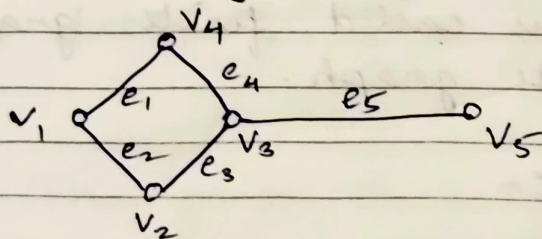
## Handshaking Theorem

\* The sum of degree of the vertices of a graph is equal to twice the no. of edges

Consider a graph  $G$  with  $e$  edges &  $n$  vertices  $V_1, V_2, \dots, V_n$ .  $\because$  each edge contributes 2 degree in a vertex, the sum of degree of all vertices will be  $2 \times e$

$$\sum_{i=1}^n d(V_i) = 2e$$

~~ie~~  
Eg: Verifying handshaking theorem for the graph.



No. of edges =  $e = 5$

$$d(V_1) = 2$$

$$d(V_2) = 2$$

$$d(V_3) = 3$$

$$d(V_4) = 2$$

$$d(V_5) = 1$$

~~Sum of diagrams~~ <sup>Sum of degree</sup> ~~= 10~~

$$\text{Sum of degree} = 2 \times e$$

$$= 2 \times 5 = 10$$



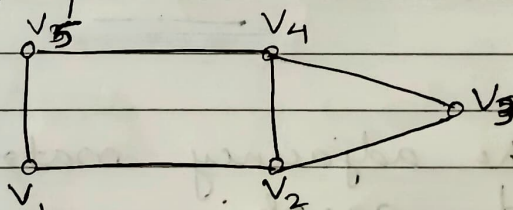
## Matrix representation of graphs

### ① Adjacency matrix:-

Let  $a_{ij}$  denotes no. of edges of  $(v_i, v_j)$  then  $A = \text{matrix}[a_{ij}]$  is called adjacency matrix

$$a_{ij} = \begin{cases} 1, & \text{if } v_i, v_j \text{ has an edge} \\ 0, & \text{otherwise.} \end{cases}$$

Q. Find the adjacency matrix of the following graph.



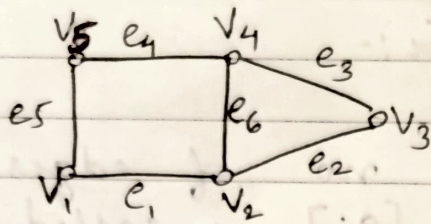
Ans:

$$A = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 & v_5 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

### ② Incidence matrix:-

Let  $G$  be a graph with  $m$  vertices,  $n$  edges then matrix  $M = [m_{ij}]$  defined by  $m_{ij} = 1$ , if  $v_i$  is incident to  $e_j$ .  
 $m_{ij} = 0$ , if  $v_i$  is not incident to  $e_j$ .  
 $m_{ij} = 2$ , if  $e_j$  is a loop.

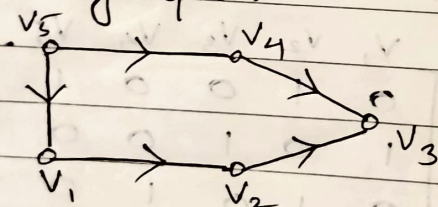
Q. Find the incident matrix of the graph given below:



Ans:

$$M = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \end{matrix} \\ \begin{matrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix} \end{matrix} \quad \begin{matrix} \text{edges touching} \\ V_i = 1 \end{matrix}$$

Q. Find the adjacency matrix of the directed graph:



Ans:

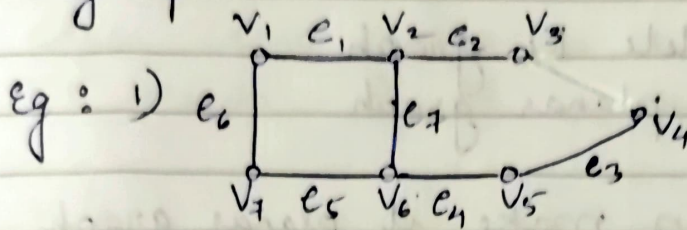
$$A = \begin{matrix} & \begin{matrix} V_1 & V_2 & V_3 & V_4 & V_5 \end{matrix} \\ \begin{matrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix} \quad \begin{matrix} \text{direction} \\ \text{noted} \end{matrix}$$

Bigraph or Bipartite graph :-

If the vertex set  $V$  of a graph  $G$  can be partitioned into 2 non-empty disjoint subsets  $X$  &  $Y$ , in such a way that



edge of  $G$  has one end in  $X$  & other end in  $Y$ , then  $G$  is called Bigraph.

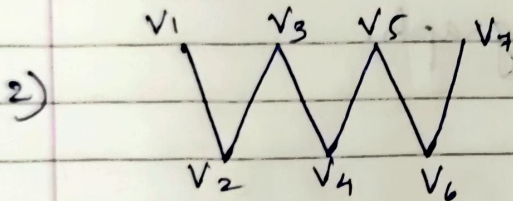


$e_1$ 's 1 vertex  $v_1$  will put in  $X$  &  $v_2$  will put in  $Y$ .

$$V = \{v_1, v_2, v_3, v_4, v_5, v_6, v_7\}$$

$$X = \{v_1, v_3, v_4, v_6\}$$

$$Y = \{v_2, v_5, v_7\}$$

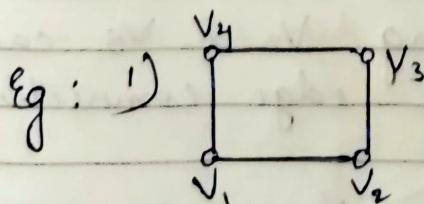


$$X = \{v_1, v_3, v_5, v_7\}$$

$$Y = \{v_2, v_4, v_6\}$$

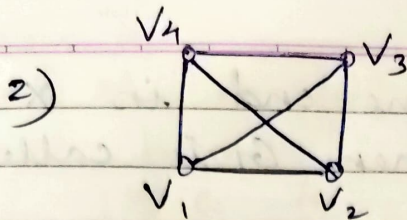
Planar Graph:-

A graph which can be drawn in a plane so that its edges do not cross is called planar graph.



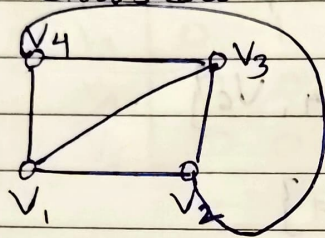
$K_4$  planar graph.



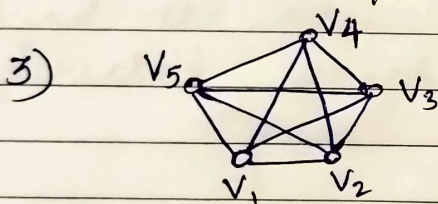


Complete  $K_4$  graph  
 Non-planar graph

We can make it planar graph  
 by drawing the edge  $V_2$  &  $V_4$   
 by outside



$K_4$  planar graph:

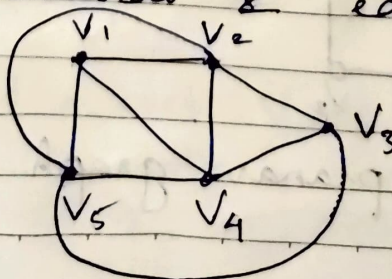


$K_5$  - complete graph  
 Non-planar graph

We cannot make it planar.

Q Show that the complete graph  $K_5$  is not planar. The edge connecting  $V_2$  &  $V_3$  can be drawn outside & edge connecting  $V_3$  &  $V_5$ .

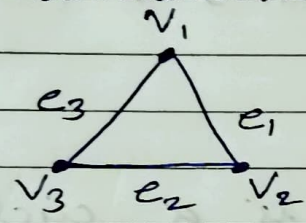
Ans: The edge connecting  $V_2$  &  $V_5$  can be drawn outside & edge connecting  $V_3$  &  $V_5$ .



But the edge ~~can~~ connecting  $V_1$  &  $V_3$  cannot be drawn outside because from outside the edges are crossed. Hence the edges cross each other in a complete  $K_5$  graph.  $\therefore$  It is not planar.

### Walk

A walk in a graph is defined as a finite sequence of vertices & edges arranged alternatively.



$V_1, e_1, V_2, e_2, V_3, e_3, V_1$

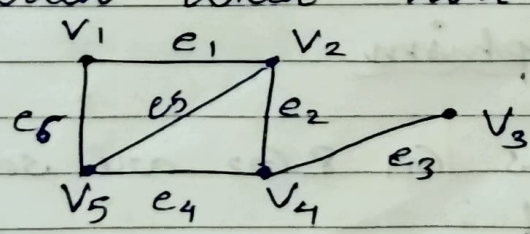
length of walk = no. of edges in a walk.

(w)  
length<sub>w</sub> = 3.

### Trial

A walk with distinct edges is called Trial.

ie a walk with non-repeated edges.



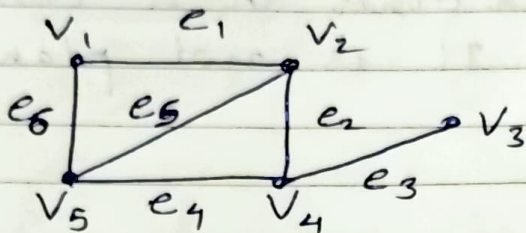
Trial =  $V_1, e_1, V_2, e_2, V_4, e_4, V_5, e_6, V_1$



## Path

Path is a walk with distinct vertices  
ie.

Eg:



- 1)  $V_1, e_1, V_2, e_5, V_4, e_6, V_1$  (cycle)
- 2)  $V_1, e_1, V_2, e_2, V_3, e_3, V_4$  (not cycle)

## Cycle

A closed path is a cycle, no edges is repeated & the starting & ending vertex are same.

## Circuit

A closed trail is a circuit. Edges are not repeated. Vertices may repeat. If all vertices are distinct from initial vertex to terminal vertex, then walk is called circuit. Every cycle is a circuit, But not every circuit is cycle.



## Isomorphism

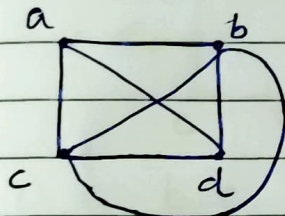
2 graphs  $G_1$  &  $G_2$  are said to be isomorphic if:

- 1) They have same no. of vertices.

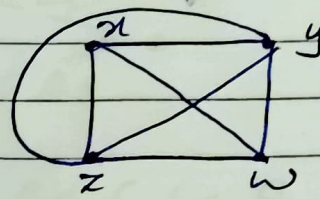
- 2) Same no. of edges.
- 3) Same degree for corresponding vertices.
- 4) Same no. of loops.

We write  $G_1 \cong G_2$

Q. Check whether the following graph are isomorphic or not.



$G_1$



$G_2$

Ans: Here,  $f(a) = w$   
 $f(b) = z$   
 $f(c) = y$   
 $f(d) = x$

Edge correspondence:-

$$\begin{aligned} \{a, b\} &= \{z, w\} \\ \{a, c\} &= \{y, w\} \\ \{c, d\} &= \{x, y\} \\ \{b, d\} &= \{x, z\} \\ \{c, b\} &= \{z, y\} \\ \{a, d\} &= \{x, w\} \end{aligned}$$

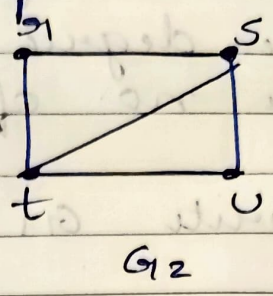
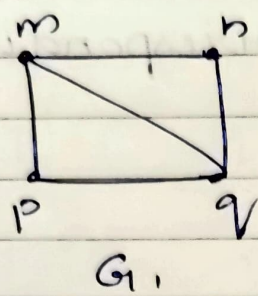
Degree:-

$$\begin{aligned} d(a) &= 3 = d(w) \\ d(b) &= 4 = d(z) \\ d(c) &= 4 = d(y) \\ d(d) &= 3 = d(x) \end{aligned}$$

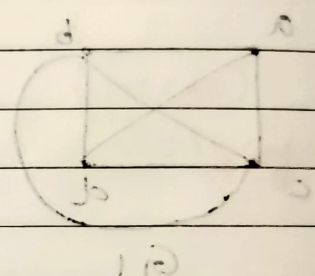
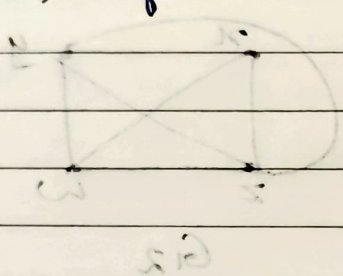
$\therefore$  Clearly  $G_1$  is isomorphic to  $G_2$ .  
 $G_1 \cong G_2$



Q. Check whether graph is isomorphic.



Ans Here,  $f(m) = s$

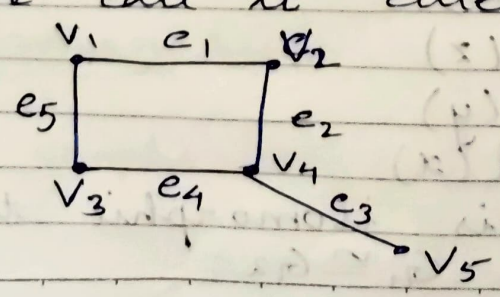


### Euler Graph

If a walk contains all the edges of the graph exactly once, then the walk is called Euler Trail.

If there is a Euler circuit in the graph we call it Euler line.

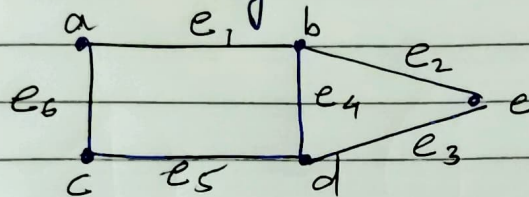
Eg:



edges cannot repeat but all edges should be there  
 $V_5 e_3 V_4 e_2 V_2 e_1 V_1 e_5 V_3 e_4 V_4$

## Hamiltonian Graph

A hameltonian graph is a simple circuit that passes through every vertex exactly once.



$a e_1 b e_2 e e_3 d e_5 c e_6 a$

every vertex  
visited.